

Essentials of Data Science Laboratory - 2304102L - 2026

Search course

1. Practical 1

1.1. Practice Lab Assignment

1.1.1. Calculate Momentum

1.1.2. Conditional Calculation Based on the...

1.1.3. Days between Two Dates

1.1.4. Reverse a Number

1.1.5. Multiplication Table

1.2. Lab Assignment

2. Practical 2

3. Practical 3

4. Practical 4

5. Practical 5

1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:
$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1
2 mass=float(input())
3 velocity=float(input())
4 momentum=mass*velocity
5 print(f"momentum: .2f)kgm/s")
```

Average time: 0.007 s

Maximum time: 0.009 s

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Actual output

5.0

5.0

10.0

10.0

50.00kgm/s

50.00kgm/s

Test case 2

Terminal

Test cases

Prev

Reset

Submit

Next

1.1.2. Conditional Calculation Based on the Number of Digits

18:15

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints: $1 \leq n \leq 999$ **Input Format:**

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

+

condition...

Submit

```
1 #write your code here...
2 import math
3 number=int(input())
4 if 1<=number<=9:
5     print(number**2)
6 elif 10<=number<=99:
7     print(f"{math.sqrt(number):.2f}")
8 elif 100<=number<=999:
9     print(f"{math.cbrt(number):.2f}")
10 else:
11     print("Invalid")
```

Terminal Test cases

< Prev Reset Submit Next >

1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`.
- The second line contains the second date in the format `YYYY-MM-DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```

1 from datetime import date
2 def is_leap(year):
3     if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
4         return True
5     return False
6
7 # Days in months
8 month_days = [31,28,31,30,31,30,31,31,30,31,30,31]
9
10 # Convert date to total days from year 0
11 def days from start(v, m, d):

```

Average time: **0.011 s** (10.60 ms) | Maximum time: **0.013 s** (13.00 ms)

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (8 ms) [Debug](#)

Expected output	Actual output
2014-07-02	2014-07-02
2014-11-02	2014-11-02
123	123

Test case 2 (10 ms)

[Terminal](#) [Test cases](#)

1.1.4. Reverse a Number 05:21

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases +

reverseN... Submit

```
1 n = int(input())
2 reversed_num = 0
3 while n > 0:
4     digit = n % 10
5     reversed_num = reversed_num * 10 + digit
6     n = n // 10
7 print(reversed_num)
8
```

Average time0.005 s4.00 ms

Maximum time0.006 s6.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 6 ms Debug

Expected output53677635

Actual output53677635

Test case 2 4 ms

Terminal

Test cases

1.1.5. Multiplication Table05:41

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases+

multipl...

```
1 #write your code here...
2 num=int(input())
3 for i in range(1,11):
4     print(f"{num} x {i} = {num*i}")
5
6
```

Average time0.006 s6.25 ms

Maximum time0.007 s7.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 17 msDebug

Expected output	Actual output
8	8
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40

Terminal

Test cases

1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:
If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and division by the number of courses.

passorFa...

Submit

1n = int(input())
2l = input()
3lst = l.split(" ")
4
5a = 0
6fail = False
7
8for i in lst:
9 if int(i) < 40:
10 print("Fail")
11 fail = True

Average time0.007 s7.40 msMaximum time0.020 s20.00 ms

5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed

Test case 17 msDebug

Expected output
4
55 65 78 89
Aggregate Percentage: 71.75
Grade: First Division

Actual output
4
55 65 78 89
Aggregate Percentage: 71.75
Grade: First Division

Test case 2Time

TerminalTest cases

PrevResetSubmitNext

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1.2.2. Fibonacci Series03:50

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

A single integer n representing the number of terms to generate.

Output Format:

A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     if n == 1:
3         return 0
4     elif n == 2:
5         return 1
6     else:
7         return fibonacci(n-1) + fibonacci(n-2)
8
9
10 n = int(input())
11 for i in range(1, n + 1):
```

Average time0.014 s14.25 ms

Maximum time0.036 s36.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 17 ms

Debug

Expected output

5

0 1 1 2 3

Actual output

5

0 1 1 2 3

Test case 27 ms

TerminalTest cases

PrevResetSubmitNext

1.2.3. Pattern - 1

05:37

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

Submit

1n=int(input())
2for i in range (1,n+1):
3print(" "*i)

Average time: 0.006 s
Maximum time: 0.011 s

2 out of 2 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case 1 11 ms Debug

Expected output: 5
Actual output: 5

Terminal Test cases

< Prev Reset Submit Next >

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1.2.4. Pattern - 201:18

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 # Type Content here...
2 n=int(input())
3 for i in range (1,n+1):
4     for j in range (1,i+1):
5         print(j,end=" ")
6     print("")
7
```

Average time0.005 s5.00 ms

Maximum time0.005 s5.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 15 msDebug

Expected output	Actual output
5	5
1 2	1 2
1 2 3	1 2 3
1 2 3 4	1 2 3 4
1 2 3 4 5	1 2 3 4 5

Terminal

Test cases

< Prev

Reset

Submit

Next >